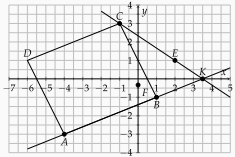
Coordinate geometry: vectors and lines

Lecture and worked examples

Parallelogram and intersection

In an orthonormal frame, $A(-4,-3)$, $B(1,-1)$ and $C(-1,3)$.

1. Determine D such that $ABCD$ is a parallelogram.
2. Determine its centre I in two ways.
3. With $E(2,1)$, determine K, the intersection of (AB) and (CE).
4. Determine F such that $\overrightarrow{AF} = \frac{2}{3}\overrightarrow{AE}$. Are D, F, B collinear?



Worked answer: Parallelogram and intersection

1. $D = A + C - B = (-6, 1)$. 2. Midpoint of $[AC]$ and of $[BD]$: $I(-5/2, 0)$. 3. $(AB) : 2x - 5y - 7 = 0$; $(CE) : 2x + 3y - 7 = 0$. Subtracting gives $8y = 0$, then $x = 7/2$, so $K(7/2, 0)$. 4. $\overrightarrow{AE} = (6, 4)$, so $F = A + (4, 8/3) = (0, -1/3)$. $\overrightarrow{DF} = (6, -4/3)$, $\overrightarrow{DB} = (7, -2)$; determinant $-12 + 28/3 = -8/3 \neq 0$. The three points are not collinear.

Comparing a vector method and a coordinate method

In a non-degenerate triangle ABC, I is the midpoint of $[AB]$ and J the midpoint of $[AC]$.

1. Prove with vectors that $(IJ) \parallel (BC)$.
2. Check with $A(0,0)$, $B(6,0)$, $C(2,4)$.
3. For these coordinates, determine the equations of (IJ) and (BC), then the ratio IJ/BC.

Worked answer: Comparing a vector method and a coordinate method

1. $\overrightarrow{IJ} = \overrightarrow{IA} + \overrightarrow{AJ} = \frac{1}{2}(\overrightarrow{BA} + \overrightarrow{AC}) = \frac{1}{2}\overrightarrow{BC}$. The vectors are nonzero and collinear.
2. $I(3,0)$, $J(1,2)$; $\overrightarrow{IJ} = (-2,2)$ and $\overrightarrow{BC} = (-4,4)$.
3. $(IJ) : x + y - 3 = 0$; $(BC) : x + y - 6 = 0$. Their direction vectors are collinear. $IJ = \sqrt{8}$, $BC = \sqrt{32}$, so $IJ/BC = 1/2$. The vector proof is general; the numerical check only proves this particular case.

Comparing a vector method and a coordinate method

A figure seems to show two parallel lines. What is the difference between this observation and a proof?

Worked answer: Comparing a vector method and a coordinate method

The figure suggests a conjecture and may be imprecise. A proof uses exact data and a property, for example a zero determinant of nonzero direction vectors. A conclusion must state the objects concerned.

Recognising a configuration and justifying

In an orthonormal frame, $A(0,0)$, $B(4,2)$, $C(3,4)$ and $D(-1,2)$.

1. Show that $ABCD$ is a parallelogram.
2. Calculate the four lengths and determine whether it is a rhombus.
3. Calculate AC^2, BD^2 and determine whether it is a rectangle.
4. Determine its centre and an equation of each diagonal.

Worked answer: Recognising a configuration and justifying

1. $\overrightarrow{AB} = (4, 2) = \overrightarrow{DC}$: it is a parallelogram. 2. $AB = CD = \sqrt{20} = 2\sqrt{5}$, $BC = DA = \sqrt{5}$; adjacent sides are not equal, so it is not a rhombus. 3. $AC^2 = 3^2 + 4^2 = 25$; $BD^2 = (-5)^2 + 0^2 = 25$. A parallelogram whose diagonals have the same length is a rectangle. One can also use $AB^2 + BC^2 = 20 + 5 = 25 = AC^2$ and the converse of Pythagoras. 4. The centre is $I(3/2, 2)$. $(AC) : 4x - 3y = 0$; $(BD) : y = 2$. Their intersection recovers I.